Colour

One of the most desirable qualities in a gemstone is beauty, and an important part of this is the stone's colour. In gems, colour is caused when light is absorbed within the crystal, or refracted – changing direction as it passes through the gem. White light is composed of many colours; when one or more of those colours is absorbed, the remaining light emerging from the gem is coloured. This can be brought about by the presence of trace elements that cause certain wavelengths to be absorbed, or can be a part of the gem's chemical structure (see below).

Idiochromatic gems Idiochromatic gems are those sometimes described as "self-coloured", as their colour is inherent in their chemical make-up. Rhodochrosite is a manganese carbonate with a naturally pink to red colour due to its manganese content, and peridot is an iron magnesium silicate, which is green as a result of its iron content.

Peridot

Ruby

Allochromatic gems Allochromatic gems are those coloured by trace elements in their structure. Amethyst and ruby are examples of these: amethyst is colourless quartz made purple by traces of iron, while ruby is corundum coloured by traces of chromium.

Parti-colouring Gems with different colours within the same stone are called parti-coloured. Gems with two colours are called bicoloured; those with three colours, tricoloured. Rarely, a dozen or more colours can occur. Divisions between the colours can be abrupt or gradual. Parti-colouring is often caused by changes in the chemical medium in which the crystal has grown.

Watermelon tourmaline

Iolite seen from the top

Iolite seen from the side

Pleochroic gems As white light passes through a gemstone, colours are absorbed differently in different directions: as a consequence, a stone can be a different colour when viewed from different angles. This effect is called pleochroism, and it can be an important aid to the identification of cut stones.

Lustre

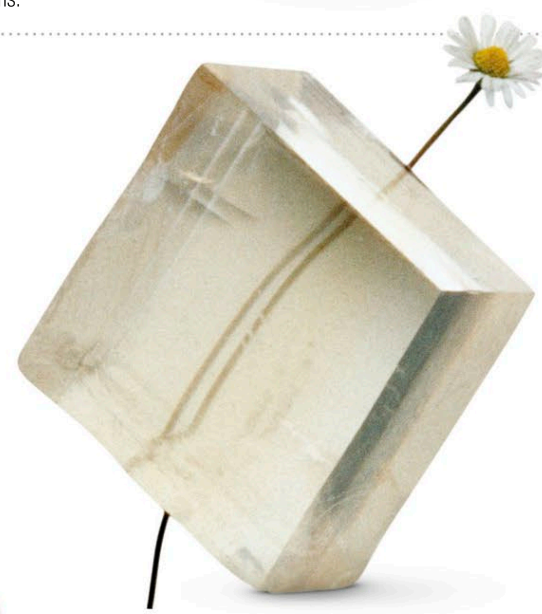
A gem's lustre is the general appearance of its surface in reflected light. There are two basic types of lustre: metallic and non-metallic. Precious metals have metallic lustres, and gemstones non-metallic, with the exception of a few like hematite and pyrite. Lustres that relate to gems include vitreous, waxy, pearly, silky, resinous, greasy, earthy, metallic, and adamantine.

Diamond

Adamantine lustre Gems that demonstrate an extraordinary brilliance and shine have an adamantine lustre. It is a relatively uncommon lustre, possessed by diamonds, some zircons, and a very few other gems.

Refractive index

When light passes through a transparent gem, it changes speed and direction. This is called refraction. The change in the speed of light as it passes from the air into a gem is called the refractive index (RI). The change in direction, or bending, of the light can be used to calculate the gem's RI. Diamond's high RI results in flashes of light seen when the gem is moved – its "fire". The greater the dispersion of the white light, the greater the fire.



High refractive index (RI)

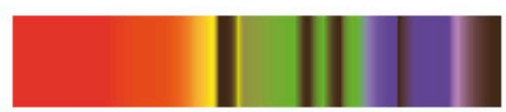
Low refractive index (RI)

Double refraction Gemstone minerals in the cubic system bend light equally in all directions. Other types of crystal system bend light in two directions, with the crystal structure causing light to bend at two different angles. This is called double refraction.

Spectroscopy

The study of the emission of light according to its wavelength is called spectroscopy. Devices known as spectroscopes are used to measure light waves as they pass through gemstones. The spectroscope has a small slit for light to pass through. When a gem is placed between a light source and the slit, a light spectrum is produced. Dark bands appear where certain wavelengths are absorbed by the stone. These bands are characteristic of various elements, enabling identification of the gem's chemical make-up. The three spectra on the right reveal much about the gems' composition.

Ruby



Almandine garnet

Red glass

Spectra Ruby's simple composition shows only a few dark bands or lines. Almandine, meanwhile, shows numerous lines, due to its complex composition, while glass is made of only two elements, so displays only two absorption areas.